

7. 有 Ω 和 $\mathbb{R}$, 定义 $f: \mathbb{T} \subseteq S \times \mathbb{R}$.

为高公理模式替换公理模式保证, 可以如此构造 $\mathcal{F}$, $\mathcal{F} = \{ (w, r) \in \mathbb{Q} \times \mathbb{R} \mid \varphi(w, r) \}$

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Chapter 18 Random Variables

Similarly, M is a function mapping each outcome another way:

$$\begin{array}{ll} M(HHH) = 1 & M(THH) = 0 \\ M(HHT) = 0 & M(THT) = 0 \\ M(HTH) = 0 & M(TTH) = 0 \\ M(HTT) = 0 & M(TTT) = 1 \end{array}$$

So C and M are random variables.

18.1.1 Indicator Random Variables

An *indicator random variable* is a random variable that maps every outcome to either 0 or 1. Indicator random variables are also called *Bernoulli variables*. The random variable M is an example. If all three coins match, then $M = 1$; otherwise, $M = 0$.

Indicator random variables are closely related to events. In particular, an indicator random variable partitions the sample space into those outcomes mapped to 1 and those outcomes mapped to 0. For example, the indicator M partitions the sample space into two blocks as follows:

$$\underbrace{HHH \ TTT}_{M=1} \quad \underbrace{HHT \ HTH \ HTT \ THH \ THT \ TTH}_{M=0}$$

In the same way, an event E partitions the sample space into those outcomes in E and those not in E . So E is naturally associated with an indicator random variable, I_E , where $I_E(\omega) = 1$ for outcomes $\omega \in E$ and $I_E(\omega) = 0$ for outcomes $\omega \notin E$. Thus, $M = I_E$ where E is the event that all three coins match.

18.1.2 Random Variables and Events

There is a strong relationship between events and more general random variables as well. A random variable that takes on several values partitions the sample space into several blocks. For example, C partitions the sample space as follows:

$$\underbrace{TTT}_{C=0} \quad \underbrace{\cancel{TTH} \quad \cancel{THT} \quad HTT}_{C=1} \quad \underbrace{TTH \quad HTH \quad HHT}_{C=2} \quad \underbrace{HHH}_{C=3}.$$

Each block is a subset of the sample space and is therefore an event. So the assertion that $C = 2$ defines the event

$$[C = 2] = \{THH, HTH, HHT\},$$

and this event has probability

$$\Pr[C = 2] = \Pr[THH] + \Pr[HTH] + \Pr[HHT] = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = 3/8.$$

2° 从集合函数 Σ -代表, 为了更清楚地说明和不可测的奇特集合, 我们引入 Σ -函数 σ 的概念, 其明确了哪些子集是可以分给概率的.

⇒ 本质仍是等面积块, 不过单位面积, 某些单位面积大的小块其实是很多个等面积块的累加。由于趋近 0 着起耳, 测度差不多, 但构造完全不一, 由变量积分定义, 给予“概率密度”!